

Ziad Eid Elsaid Cairo, Egypt | 01140012767 | zyadeid28@gmail.com |

LinkedIn: [linkedin.com/in/ziad-eid-66b960231/](https://www.linkedin.com/in/ziad-eid-66b960231/)

Summary

Highly motivated R&D Software Engineer bridging the gap between embedded systems hardware and high-level software development, specializing in medical device innovation. Skilled in embedded system design, software-hardware integration, 3D medical imaging, and microcontroller programming. Dedicated to building reliable, safety-compliant solutions that advance healthcare technology.

Experience

Elezaby Medical Factory | Cairo, Egypt *R&D Software Engineer* | 2024 – Present

- Built a prototype 3D medical imaging system simulating CBCT data processing workflows, bridging hardware outputs with high-level software visualization.
- Transformed raw CT slice datasets into volumetric data structures and 3D anatomical models, utilizing segmentation techniques to isolate bone structures.
- Implemented 3D reconstruction and mesh generation pipelines for interactive visualization, optimizing processing times for cross-platform environments (Windows/Linux).
- Developed and programmed embedded control systems for critical medical devices (autoclaves, warming cabinets, heat sealing machines) using ATmega32 and Atmel Studio.
- Designed modular embedded software architecture for multi-cycle sterilization control, reducing potential system failures through real-time monitoring and safety interlocks.
- Conducted rigorous system simulation and testing using Proteus prior to hardware implementation, collaborating with engineering teams in an Agile environment for rapid prototyping.

Internships & Training

Harvard University (Online) *Introduction to Data Science with Python* | 2026

- Applied Python for modeling, statistics, and machine learning utilizing Pandas, NumPy, Matplotlib, and scikit-learn to solve real-world data challenges.

Information Technology Institute (ITI) | Smart Village, Egypt *Software Quality Assurance Tester* | Aug 2024 – Sep 2024

- Applied manual testing and debugging techniques for embedded software systems within the Software Testing Life Cycle (STLC).

ITI Knowledge City *Embedded Systems Diploma* | Jul 2024 – Aug 2024

- Developed hands-on projects focusing on C programming, AVR microcontrollers, communication protocols (I2C/SPI/UART), and embedded system testing.

Japanese for Tools and Molds Manufacturing | 10th of Ramadan City, Egypt *Mechanical Design Intern* | Aug 2023 – Sep 2023

- Designed mechanical components and assemblies using SOLIDWORKS for mold manufacturing.

Siemens Energy | Ain Sokhna, Egypt *Operations of Digital Circuits Intern* | Apr 2023 – May 2023

- Tested, troubleshoot, and designed digital circuits, gaining practical experience in signal processing.

BEDO Innovating Education | 6th of October City, Egypt *Electrical Hydraulic System Intern* | Feb 2022

- Assembled, tested, and troubleshoot electro-hydraulic systems and electrical components.

Academic Projects

- **EEG-Based Prosthetic Arm:** Designed an EEG-based control system for prosthetic arm movement. Presented at Made in Egypt (MIE), Technology Universities Hackathon, and Startup Olympics.
- **Obstacle Avoiding Car & Mine Detector:** Implemented ultrasonic-based obstacle detection using Arduino and C programming.

Competitions & Leadership

- Team Leader, Embedded Systems Team – EVER Electric Vehicle Rally
- Project Presenter/Competitor (EEG-Based Prosthetic Arm) – Made in Egypt (MIE), Technology Universities Hackathon, and Startup Olympics

Education

New Cairo Technological University | Egypt *B.Sc. in Mechatronics Engineering* | 2024

Technical Skills

- **Medical Technology:** CBCT Simulation, 3D Reconstruction, Sterilization Control Systems, Autoclaves, Warming Cabinets, Incubators, Ventilators
- **Programming & Tools:** C, Embedded C, Python, C++, Atmel Studio, Eclipse, VS Code, Proteus, Arduino IDE
- **Embedded Systems:** ATmega32, Arduino, I2C/SPI/UART, Sensors & Actuators, Real-Time Control Systems, Firmware Testing
- **Software & Data:** Modular Programming, Data Structures, Pandas, NumPy, scikit-learn, Agile Methodology
- **Hardware & Mechanical:** SOLIDWORKS, PCB Design, Circuit Debugging, Power Electronics, Signal Conditioning

Languages

- Arabic (Native)
- English (Fluent)
- German (A1.2)